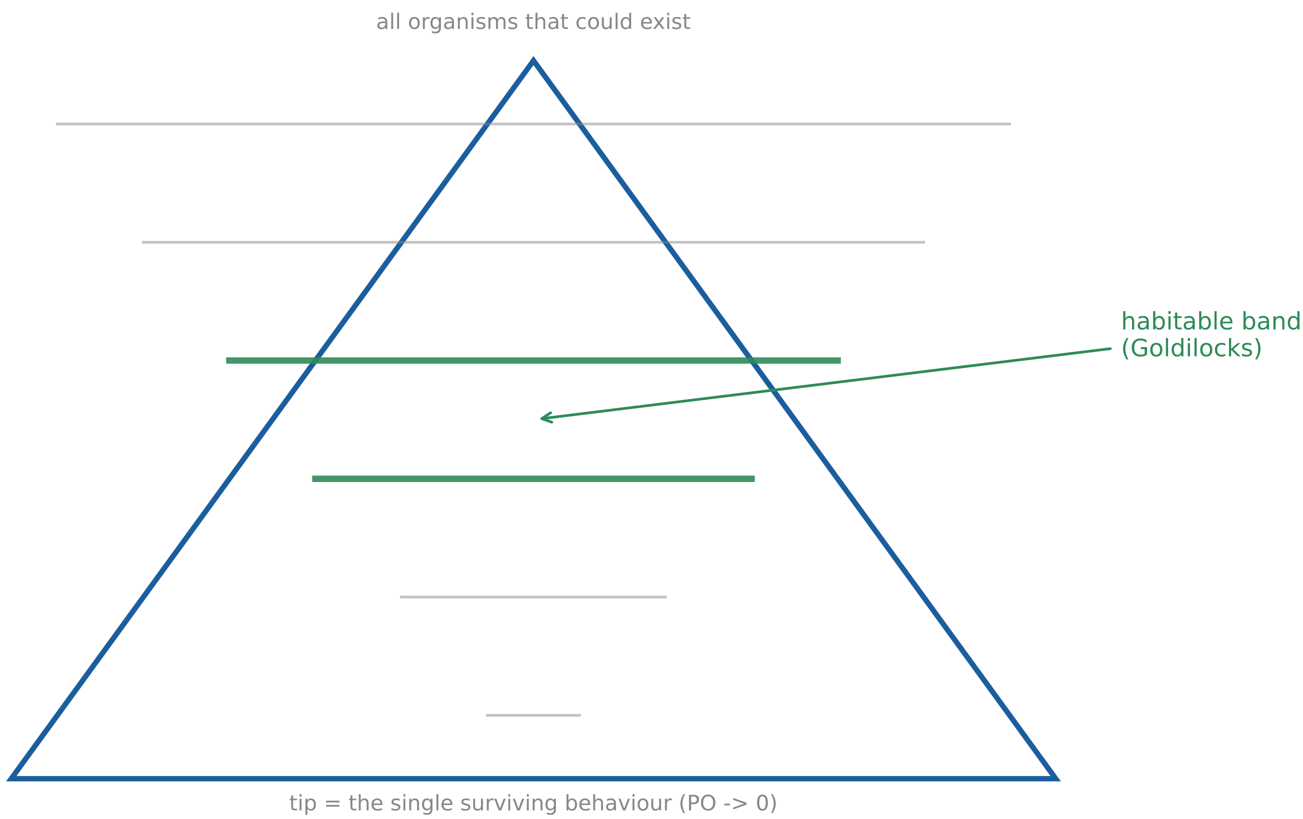
EEC — Constructing Organisms by Constraint

Findings report: laws of existence, the degradation budget,
and PO as the native complexity metric



Gradient-free neuroevolution · recurrent organism · read the STATE, not the output

GENREG / EEC · 2026-06-20

The PO / Fitness coordinate

PO (Practical Optimization) is the native complexity unit of an evolutionary model: not parameters, but CONSTRAINTS — how many laws of existence are required to collapse the infinite cone of possible organisms down to the behaviour we need. PO counts AXES covered, not constraints stacked (two constraints on one axis are redundant). Fitness is the organism's own view: how well it survives the laws imposed. PO measures top-down (how much of infinity eliminated); fitness measures bottom-up (how well it lives under what remains). They are inverse twins of the same convergence, approaching the same unreachable tip from opposite sides.

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organism identity  =  ( PO ,  fitness )
PO  -> 0      as more axes are covered   (space collapses to the survivor)
fitness -> 100  as every imposed law is solved
( PO = 0 )  <=>  ( fitness = 100 )   — asymptotes, never reached
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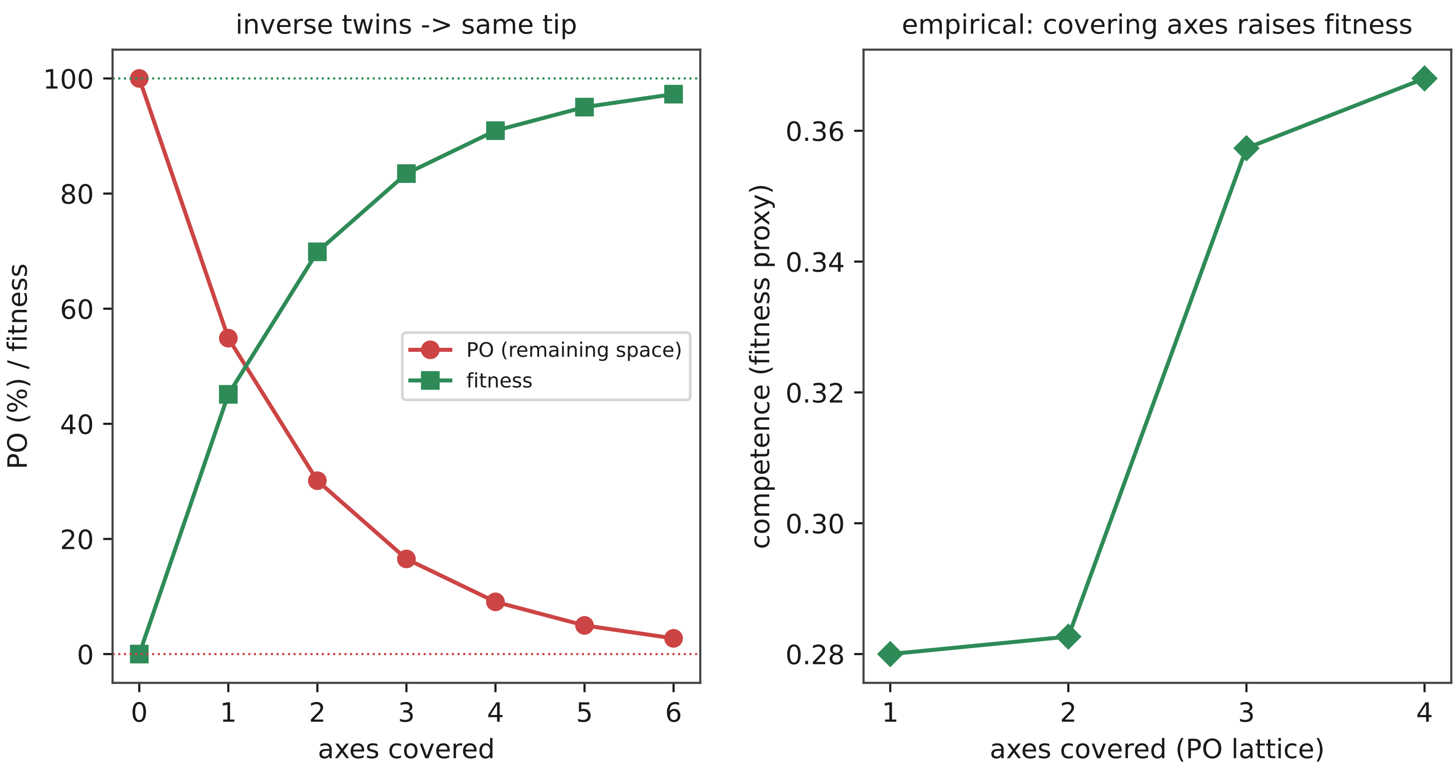

Left: the law, idealized. Right: measured — mean competence rises as more (needed) axes are covered, until conflicting axes stall it.

Constraint catalog (laws of existence)

Law	Axis	Verdict	Key result
ENERGY	survival	REAL	base law; lifespan = fitness
TIME / Occam	parsimony	REAL	selects generalization
MEMORY-RENT	parsimony	REAL	M evolves against energy
ENTROPY (decay)	maintenance	REAL	recurrent gain x3.2
OCCLUSION	observation	REAL*	gain 0.06->0.37, Goldilocks peak rho~0.4
NOISE	observation	WEAK	gain 0.06->0.14; missing >> corrupted
SCARCITY	diversity	GATED	text niches 1.5->7.5; inert long-range
REPRODUCTION-COST	selection	GATED	surplus +41%, Gini 0->0.11; text starves
MORTALITY	survival (alt)	SWAP	viable alone (0.65>0.54); redundant on energy
PERCEPTION-COST	attention	WALL	economy yes (5x); selective attn no
NON-STATIONARITY	adaptability	REAL	diversity 1.3->2.0; recovers 84%

Verdicts:

REAL = real & productive   GATED = fires only where the world pays   WEAK = present but small

WALL = search-walled (needs a channel, not pressure)   SWAP = valid alternative on the survival axis

* OCCLUSION shows a Goldilocks band (peaks then collapses).

Interactions:

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occ x entropy  (same axis)      -> INTERFERE  (combined 0.21 < either 0.37 / 0.54)
occ x repro-cost (diff axes)    -> COMPOUND   (combined 0.48 > both)
parsimony x memory (conflict)   -> SUBTRACT   (gain 0.36 -> 0.13 when stacked)
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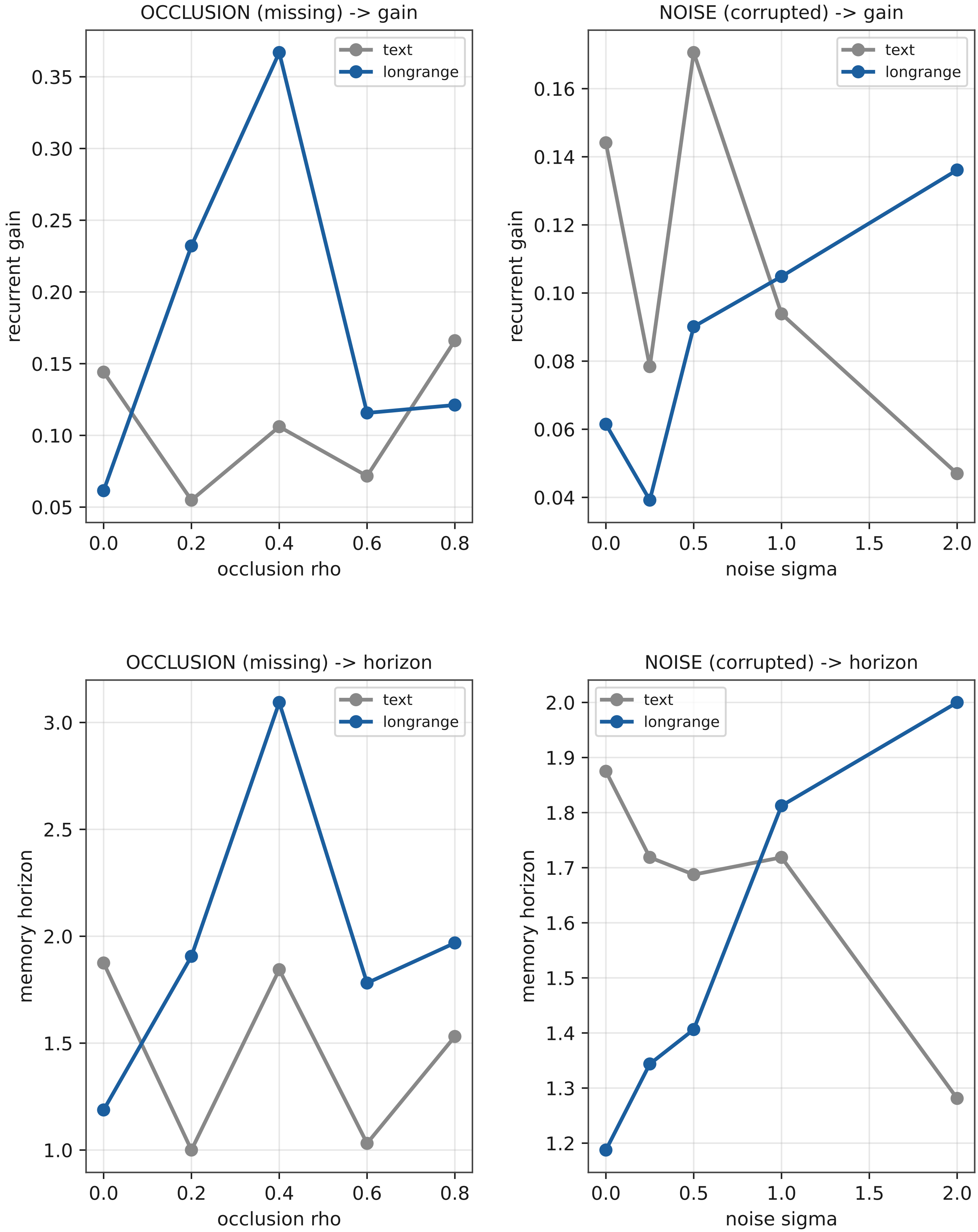
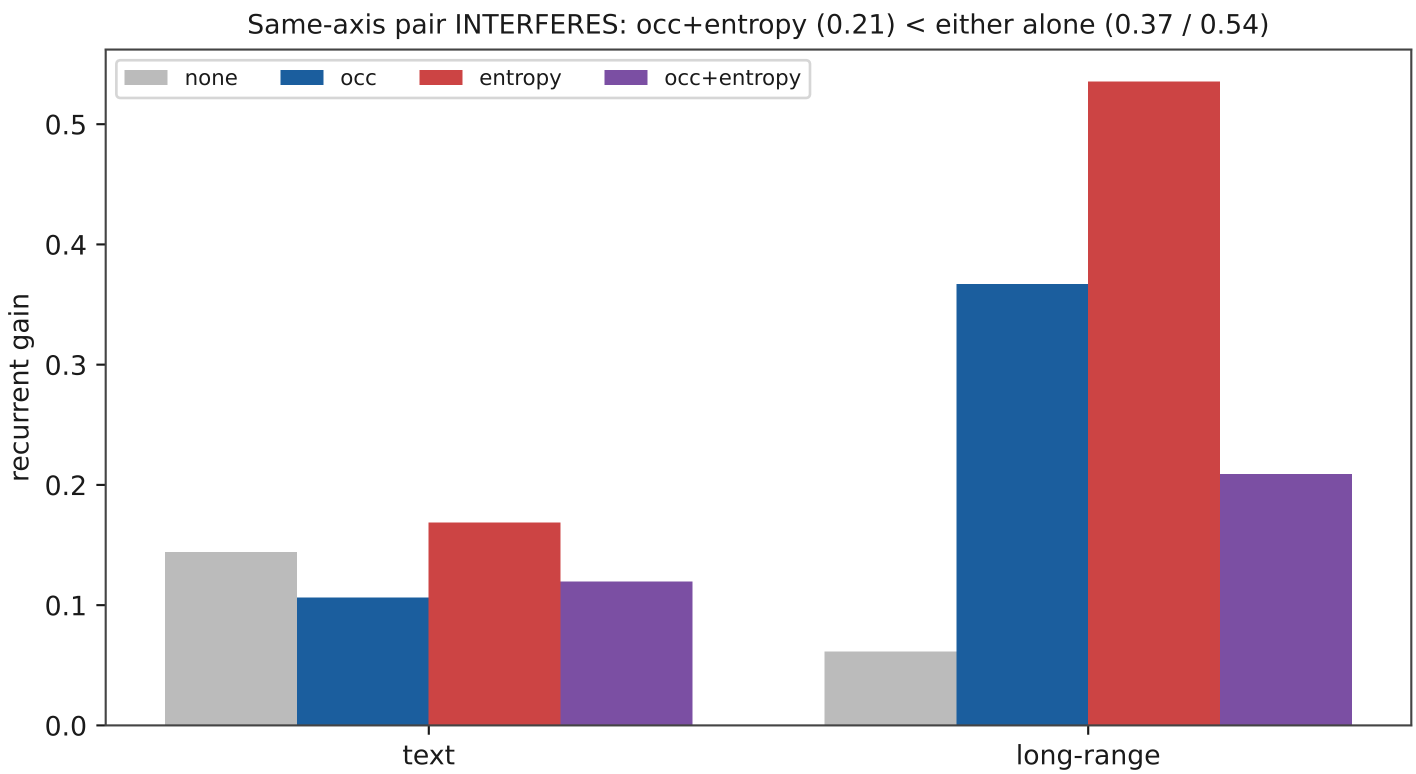
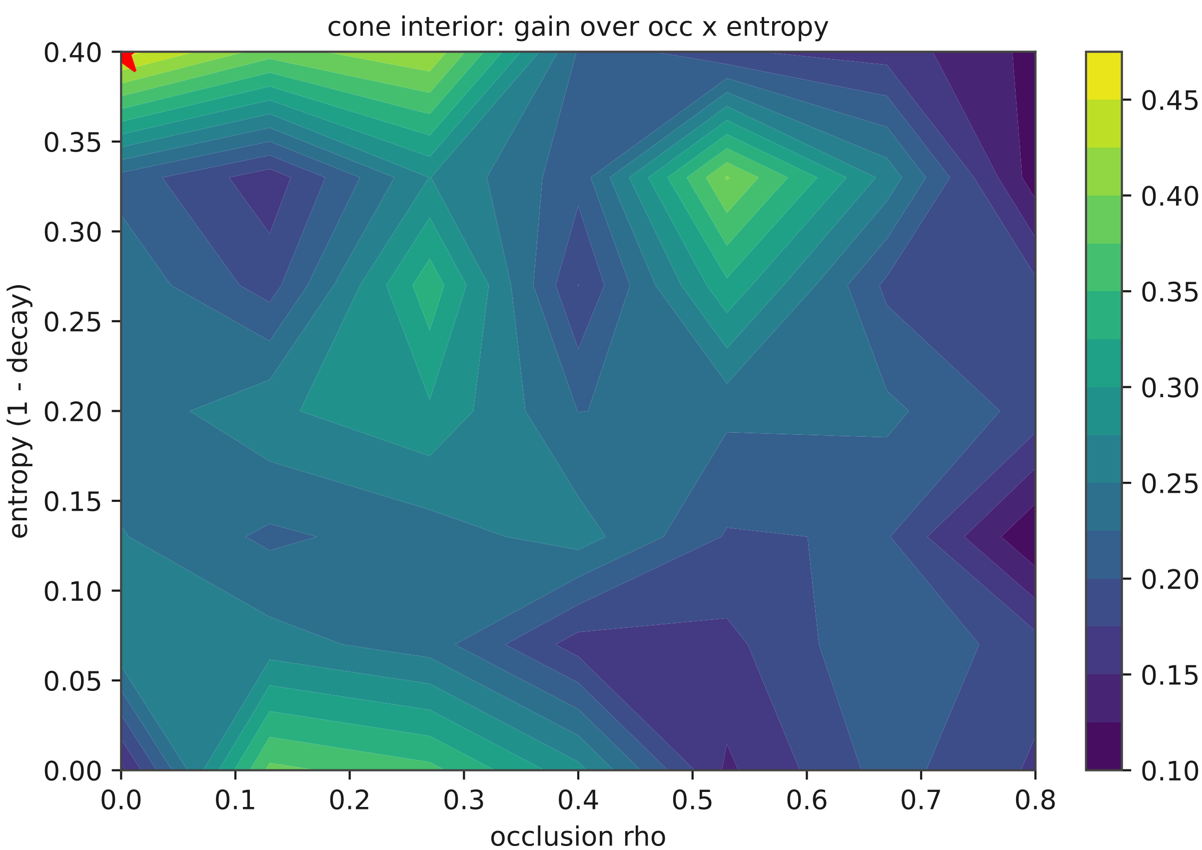
Observation axis: occlusion vs noise (memory machinery)



n drives memory in long-range (Goldilocks peak ~0.4); noise is far weaker; text (thin signal) is inert. 'Missing' forces memory; 'corrupted' only n

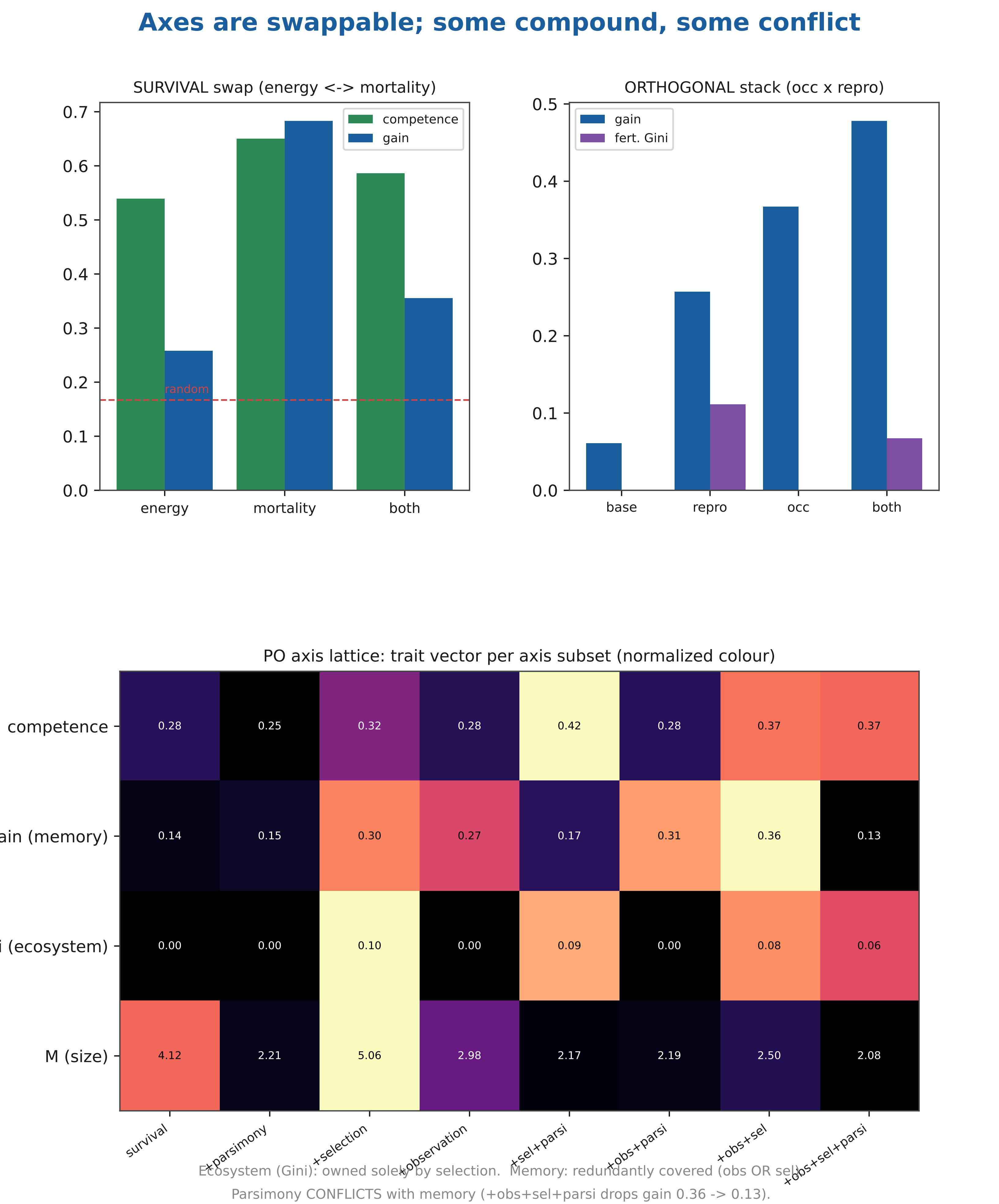
Degradation budget: interference & the cone interior



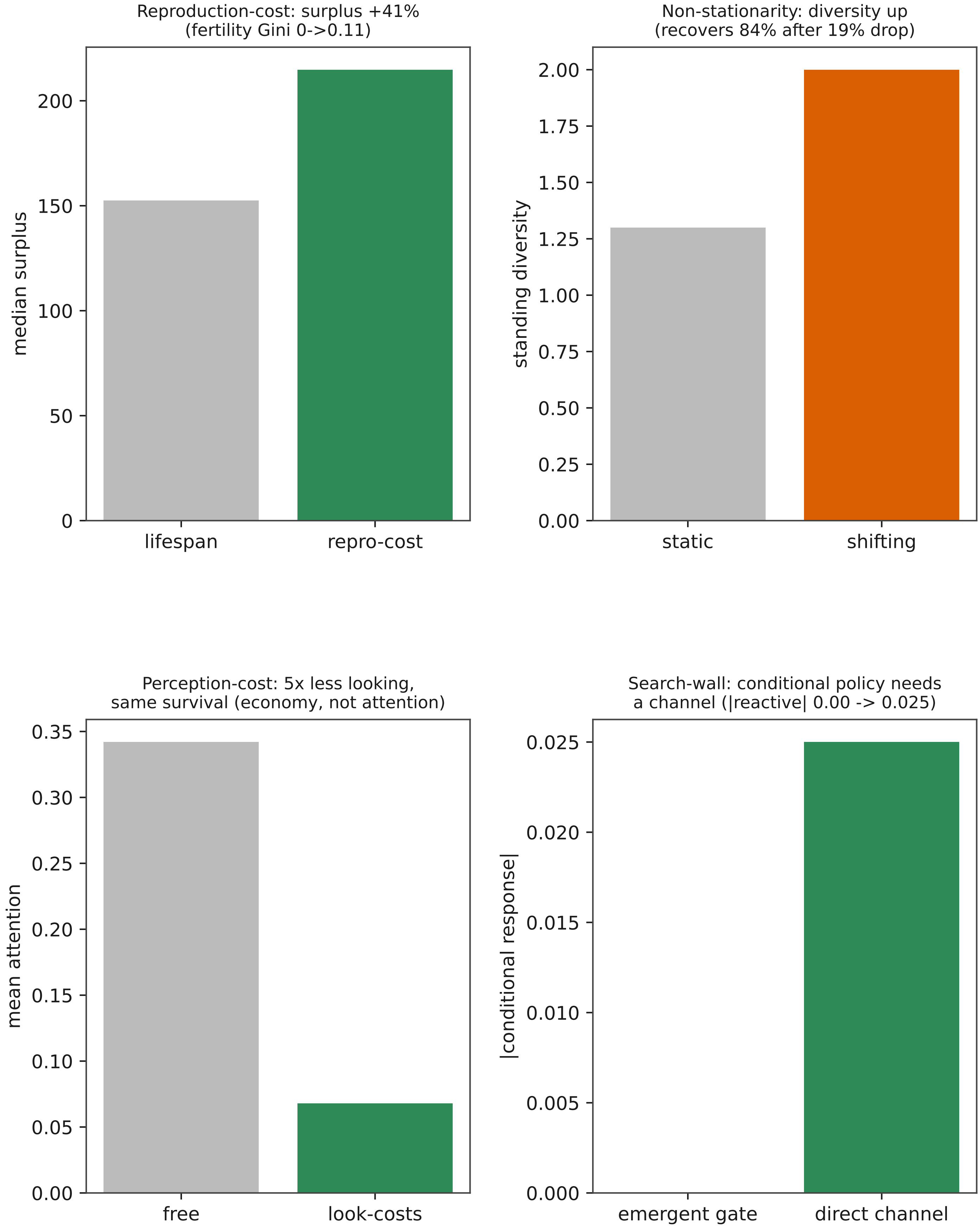


No synergistic interior ridge: the optimum is single-axis (best edge > best interior).

Co-maxing both -> collapse valley (top-right). Constraints share a finite coping budget.


Axes are swappable; some compound, some conflict
SURVIVAL swap (energy <-> mortality)
competence
gain
random
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
energy
mortality
both
ORTHOGONAL stack (occ x repro)
gain
fert. Gini
0.0
0.1
0.2
0.3
0.4
0.5
base
repro
occ
both
PO axis lattice: trait vector per axis subset (normalized colour)
competence
ain (memory)
i (ecosystem)
M (size)
survival
+parsimony
+selection
+observation
+sel+parsi
+obs+parsi
+obs+sel
+obs+sel+parsi
0.28 0.25 0.32 0.28 0.42 0.28 0.37 0.37
0.14 0.15 0.30 0.27 0.17 0.31 0.36 0.13
0.00 0.00 0.10 0.00 0.09 0.00 0.08 0.06
4.12 2.21 5.06 2.98 2.17 2.19 2.50 2.08
Ecosystem (Gini): owned solely by selection. Memory: redundantly covered (obs OR sel).
Parsimony CONFLICTS with memory (+obs+sel+parsi drops gain 0.36 -> 0.13).

Further laws: reproduction, adaptability, perception, search-wall



n law fires only where the world makes its capability pay AND reachable; conditional policies need a directly-searchable channel, not more press

Synthesis

Meta-principles (the laws about laws)

P1 Reachability: a capability emerges only where the world makes it pay AND the organism can reach the alternative by mutation. Pressure shapes how well a reachable solution is found; it does not make unreachable solutions reachable.

P2 Degradation budget / wall thickness: same-axis constraints share a finite coping budget. An outer boundary (too little pressure) and an inner boundary (too much -> unlearnable) bound a habitable Goldilocks band. The cone has thickness; the organism lives in the band, not at the tip.

P3 Two ropes: same capability label, different internal mechanics (entropy drives gain; occlusion drives horizon). Read multiple internal facets, never one metric.

P4 Orthogonal axes stack cleanly (even synergize); same-axis pairs interfere; cost-vs-capability pairs can outright conflict (parsimony subtracts memory).

P5 Search-wall: conditional/policy capability is unreachable by mutation when a constant suffices - it needs a directly-searchable channel, not more pressure.

P6 The reproduction operator is itself a law: turnover dynamics are load-bearing.

P7 Axes, not constraints. Cover each needed axis once, with the instantiation the world favors, avoiding axes that conflict with desired traits. PO = axes covered. (PO, fitness) is the full coordinate.

Method

Read the STATE, not the output - headline numbers were repeatedly selection-collapse artifacts. Beware survival-saturation (tighten energy so survival depends on the capability under test). Verify before celebrating: two independent lines agreeing is what makes a finding trustworthy.

Open frontier

Not every axis has a symmetric swap (observation is occlusion-dominant across worlds tested). A genuine noise-favoring world would require a distributed-computation task (target = function of many noisy inputs, nothing worth holding) - deliberately not built, to avoid Goodhart-ing the test.